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Inventor(s)	Hung; Chi-Yu et al.

Electronic device including a movably connected bezel

Abstract

An electronic device is provided, and the electronic device includes a base, a cover plate, and a bezel. The base has a sidewall where an opening portion is formed. An airflow selectively passes through the opening portion. The cover plate is pivotally connected to the base. The bezel is movably connected to the cover plate, wherein the bezel is rotated facing the opening portion selectively. The arrangement of the bezel may reduce the gap between the bezel and the down edge of the opening portion of the base, reducing the airflow flowing downward back to the heat-dissipation mechanism in the base. Therefore, the heat-dissipation efficiency of the electronic device is enhanced.

Inventors: Hung; Chi-Yu (New Taipei, TW), Wu; Chung-Jung (New Taipei, TW), Zeng; Ying-Sheng (New Taipei, TW)

Applicant: Wistron Corp. (New Taipei, TW)

Family ID: 1000008821500

Assignee: Wistron Corp. (New Taipei, TW)

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Primary Examiner: Parker; Allen L

Assistant Examiner: Krim; Peter

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of Taiwan Patent Application No. 111136880, filed on Sep. 29, 2022, the entirety of which is incorporated by reference herein.

BACKGROUND

Technical Field

(2) The disclosure relates to an electronic device, and in particular to an electronic device that includes a bezel.

Description of the Related Art

(3) With the advancements being made in modern technology, large electronic devices have found wider application. Portable electronic devices, such as computers and cell phones in particular, have gradually become indispensable parts of daily life. The computing ability of these electronic devices is improving. However, a large amount of heat is generated during the operation of these electronic devices. How to improve the heat dissipation efficiency of electronic devices, and

thereby to maintain the level of performance of said electronic devices, is a very important issue.

BRIEF SUMMARY

(4) The present disclosure provides an electronic device. The electronic device includes a base, a cover plate, and a bezel. The base has a sidewall, wherein an opening portion is formed on the sidewall, and an airflow selectively passes through the opening portion. The cover plate is pivotally connected to the base. The bezel is movably connected to the cover plate, wherein the bezel is rotated to face the opening portion selectively.

(5) The present disclosure provides an electronic device. The electronic device includes a base, a cover plate, and a bezel. The base has a sidewall. An opening portion is formed on the sidewall. An airflow selectively passes through the opening portion. The cover plate is pivotally connected to the base. The bezel is connected to the base and the cover plate. The bezel is disposed below the opening portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure may be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

(2) FIG. 1 illustrates a perspective view of an electronic device in accordance with some embodiments of the present disclosure.

(3) FIG. 2 illustrates an exploded view of the electronic device in accordance with some embodiments of the present disclosure.

(4) FIG. 3 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

(5) FIG. 4 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

(6) FIG. 5 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

(7) FIG. 6 illustrates a perspective view of the interior structure of the bezel in accordance with some embodiments of the present disclosure.

(8) FIG. 7 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

(9) FIG. 8 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

(10) FIG. 9 illustrates a cross-sectional view of the electronic device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(11) The tray and the electronic device of some embodiments of the present disclosure are described in the following description. However, it should be appreciated that the following detailed description of some embodiments of the disclosure provides various concepts of the present disclosure which may be performed in specific backgrounds that may vary widely. The specific embodiments disclosed are provided merely to clearly describe the usage of the present disclosure by some specific methods without limiting the scope of the present disclosure.

(12) In addition, relative terms such as “down” or “bottom,” “up” or “top” may be used in the following embodiments in order to describe the relationship between one element and another element in the figures. It should be appreciated that if the device shown in the figures is flipped upside-down, the element located on the “down” side may become the element located on the “up” side.

(13) It should be understood that although the terms “first,” “second,” etc. may be used herein to

describe various elements, materials and/or portions, these elements, materials and/or portions are not limited by the above terms. These terms merely serve to distinguish different elements, materials and/or portions. Therefore, a first element, material and/or portion may be referred to as a second element, material and/or portion without departing from the teaching of some embodiments in the present disclosure. Unless defined otherwise, the first or second element, material and/or portion in the claims may be interpreted as any element, material and/or portion in the specification as long as it conforms to the description in the claims.

(14) Unless defined otherwise, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the present disclosure belongs. It should be appreciated that, in each case, the term, which is defined in a commonly used dictionary, should be interpreted as having a meaning that conforms to the relative skills of the present disclosure and the background or the context of the present disclosure, and should not be interpreted in an idealized or overly formal manner unless so defined in the present disclosure. In addition, the terms “substantially,” “approximately” or “about” may also be recited in the present disclosure, and these terms are intended to encompass situations or ranges that is substantially or exact the same as the description herein. It should be noted that unless defined specifically, even if the above terms are not recited in the description, it should be read as the same meaning as those approximate terms are recited.

(15) First, referring to FIG. 1, FIG. 1 illustrates a perspective view of an electronic device **100** in accordance with some embodiments of the present disclosure. In some embodiments, the electronic device **100** may include a cover plate **110** and a base **120**. The cover plate **110** is pivotally connected to the base **120** and rotates around the pivot point between the cover plate **110** and the base **120**, so that the user can adjust the cover plate **110** to a suitable angle for use. For example, the cover plate **110** may include a display panel that is used to display information to the user. The base **120** may have a plurality of opening portions **125** for setting up an input device (not shown). Accordingly, the user can input commands to the electronic device **100** via the above-mentioned input device to control the electronic device **100**. It should be understood that the above configuration of the cover plate **110** and the base **120** merely serves as an example, and those skilled in the art are able to arbitrarily dispose the mutually rotatable cover plate **110** and the base **120** according to the content of the present disclosure. For example, the electronic device **100** may be a notebook computer, but the present disclosure is not limited thereto. The electronic device **100** may be any electronic device that has a cover plate **110** and a base **120**.

(16) FIG. 2 illustrates an exploded view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 2, the electronic device **100** also includes a push rod **130**, a cam **140** and a bezel **150**. The cam **140** is disposed on the base **120**, the bezel **150** is movably disposed on the cover plate **110**, and the push rod **130** is connected between the cam **140** and the bezel **150**. In some embodiments, the cover plate **110** and the base **120** are connected to each other via a hinge (not shown) and are rotatable relative to each other. The cam **140** may be connected to and linked-up with the aforementioned hinge. In some embodiments, the bezel **150** is pivotally connected to the cover plate **110** and selectively pivotally rotates relative to the cover plate **110**. For example, the bezel **150** may have a linkage portion **155**, and a pivot rod **165** may be disposed to pass through the linkage portion **155** of the bezel **150** and a corresponding structure of the cover plate **110** (for example, the linkage portion **115** shown in FIG. 6). As a result, the bezel **150** may be pivotally connected to the cover plate **110** via the pivot rod **165**, thereby the range of the rotation angle of the bezel **150** may also be defined, so that the bezel **150** may be kept in a proper position.

(17) In some embodiments, the electronic device **100** may include a sensor **181** that is disposed in the cover plate **110** and another sensor **182** that is disposed in the base **120**. The sensor **181** and the sensor **182** selectively detect the relative position of the cover plate **110** and the base **120**. It should be understood that the sensor **181** and the sensor **182** are merely examples, and the sensor **181**

and/or the sensor **182** may be disposed at other positions on the cover plate **110** or the base **120**, or various types of sensors may be adopted. These configurations are all included within the scope of the present disclosure, and all possibilities will not be listed below.

(18) More specifically, when the user opens the cover plate **110** (that is, the cover plate **110** rotates relative to the base **120**), the hinge connecting the cover plate **110** and the base **120** would rotate, driving the cam **140** to rotate at the same time. The push rod **130** is connected to the cam **140** and is abutted against the bezel **150**. In some embodiments, the push rod **130** rotates along with the rotation of the cam **140**, thereby pushing the bezel **150** to rotate relative to the cover plate **110**. The operation of the push rod **130**, the cam **140** and the bezel **150** will be further explained in the following paragraphs with reference to FIGS. 3 to 5.

(19) FIG. 3 is a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 3, the cover plate **110** of the electronic device **100** is in a closed state relative to the base **120**. To be more specific, the cover plate **110** may substantially cover the top surface **122** of the base **120** (which may be defined as an angle of 0° between the cover plate **110** and the base **120**). The base **120** has a sidewall **121** that is connected to the top surface **122**. For example, the sidewalls **121** may be inclined relative to the top surface **122**. In some embodiments, the sidewalls **121** may be substantially perpendicular to the top surface **122**. An opening portion **123** is formed on the sidewall **121**. In some embodiments, a heat dissipation mechanism (not shown) may be disposed within the base **120**, and selectively inhales cold air from the outside for heat exchange, and discharges the hot air after the heat exchange through the opening portion **123** (for example, shown as the airflow F in FIG. 5).

(20) Furthermore, the cam **140** has a plurality of teeth **141**. An engaging portion **131** and an abutting portion **132** are respectively disposed at two ends of the push rod **130**. The engaging portion **131** has a plurality of teeth **131A** that are selectively coupled to the teeth **141** of the cam **140**. In some embodiments, the engaging portion **131** has a fan-shaped gear structure. In some embodiments, the numbers of teeth of the cam **140** and the engaging portion **131** are different. For example, the number of teeth of the cam **140** is smaller than the number of teeth of the engaging portion **131**, so that the bezel **150** is rotatable to a position corresponding to the closing or opening portion of the cover plate **110**. When the cover plate **110** is in the closed state, the bezel **150** may be retracted over the cover plate **110** and located above the top surface **122** of the base **120**.

(21) FIG. 4 illustrates a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 4, the user may apply force to the cover plate **110** for rotating the cover plate **110** relative to the base **120**. In the present embodiment, the angle between the cover plate **110** and the base **120** is about 45° . When the cover plate **110** rotates, the hinge and the cam **140** are driven to rotate in the direction A (for example, counterclockwise). Meanwhile, the push rod **130** combined with the cam **140** rotates in the direction B (for example, clockwise), and the abutting portion of the push rod **130** pushes the bezel **150** in the direction C, so that the bezel **150** connected to the cover plate **110** rotates toward opening portion **123** (for example, counterclockwise).

(22) FIG. 5 illustrates a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 5, the user may continue to apply force to the cover plate **110**, so that the angle between the cover plate **110** and the base **120** is about 90° (which may be defined as the cover plate **110** being in an open state relative to the base **120**). At this time, the bezel **150** connected to the cover plate **110** rotates to be positioned and facing the opening portion **123**. In some embodiments, when the cover plate **110** is opened relative to the base **120**, a first gap GA is formed between the bezel **150** and the up edge **123A** of the opening portion **123**, and a second gap GB is formed between the bezel **150** and the down edge **123B** of the opening portion **123**. For example, the first gap GA and the second gap GB may be measured in the horizontal direction (for example, the direction that is substantially parallel to the Y-axis). In some embodiments, the first gap GA is greater than the second gap GB. For example, the first gap GA

may be between about 6 mm and about 7 mm, and the second gap GB may be between about 1 mm and about 2 mm. In some embodiments, the first gap GA is greater than five times the second gap GB. The arrangement of the bezel **150** may reduce the second gap GB between the bezel **150** and the down edge **123B** of the opening portion **123**, reducing the airflow F after the heat exchange flows downward and back to the heat dissipation mechanism (not shown) in the base **120**.

Accordingly, the heat dissipation efficiency of the electronic device **100** may be improved.

(23) In some embodiments, when the cover plate **110** is opened relative to the base **120**, the up edge **151** of the bezel **150** is higher than the up edge **123A** of the opening portion **123**, and the down edge **152** of the bezel **150** is down than the down edge **123B** of the opening portion **123**. In other words, when the cover plate **110** is opened relative to the base **120**, the distance between the up edge **151** of the bezel **150** and a horizontal plane M (for example, the X-Y plane below the base **120**) is greater than the distance between the up edge **123A** of the opening portion **123** and the horizontal plane M, and the distance between the down edge **152** of the bezel **150** and the horizontal plane M is smaller than the distance between the down edge **123B** of the opening portion **123** and the horizontal plane M. As a result, it is ensured that the arrangement of the bezel **150** reduces the distance (for example, the second gap GB) between the bezel **150** and the down edge **123B** of the opening portion **123**, so as to improve the heat dissipation efficiency of the electronic device **100**. It should be noted that the above configuration is only an example, and is not intended to limit the present disclosure. For example, the angle between the cover plate **110** and the base **120** may be any angle from 0° to 90°, or even an obtuse angle greater than 90° may be formed between the cover plate **110** and the base **120**. It should be understood that the cam **140** and the push rod **130** of the present disclosure may serve as an exemplary configuration of the driving assembly for pushing the bezel **150** to rotate, but the present disclosure is not limited thereto. Any driving assembly that pushes the bezel **150** to rotate is included within the scope of the present disclosure.

(24) FIG. 6 is a perspective view of the internal structure of the cover plate **110** according to some embodiments of the present disclosure. In some embodiments, the cover plate **110** may have a connecting portion **115**, the bezel **150** may have a linkage portion **155**, and the pivot rod **165** may be disposed to pass through the connecting portion **115** of the cover plate **110** and the linkage portion **155** of the bezel **150**. In this way, the bezel **150** may be pivotally connected to the cover plate **110** via the pivot rod **165**. As shown in FIG. 6, in this embodiment, the cam **140** and the push rod **130** may be omitted, and a motor **160** is disposed on the cover plate **110**. The motor **160** selectively connects and drives the pivot rod **165** to drive the linkage portion **155** of the bezel **150** to rotate, so that the bezel **150** faces the opening portion **123** (for example, referring to FIGS. 3 to 5). For example, the electronic device **100** may have a sensor (such as the sensors **181** and **182** shown in FIG. 2), which selectively detects the relative position of the cover plate **110** and the base **120**, and the sensor may control the motor **160** (for example, when it detects that the cover plate **110** is opened) to drive the bezel **150** to rotate. It should be understood that the motor **160** and the sensor (the sensors **181** and **182**) of the present disclosure may serve as an exemplary configuration of the driving assembly for pushing the bezel **150** to rotate, but the present disclosure is not limited thereto. Any driving assembly that pushes the bezel **150** to rotate is included within the scope of the present disclosure.

(25) FIG. 7 illustrates a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 7, when viewed in a direction (such as the X direction) that is parallel to the rotation axis R (for example, the extending direction of the pivot rod **165**) of the bezel **150**, the motor **160** (shown in dotted lines) may overlap with the pivot rod **165**. In some embodiments, the motor **160** may be disposed adjacent to the connecting portion **155** of the bezel **150**. In some embodiments, the motor **160** may selectively adjust the position of the bezel **150** steplessly within the rotation range of the bezel **150**, and may precisely control the bezel **150** as required.

(26) FIG. 8 illustrates a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 8, in this embodiment, a bezel **170** may be disposed to replace the bezel **150**. In some embodiments, the first end **170A** and the second end **170B** of the bezel **170** may be connected to the cover plate **110** and the base **120**, respectively. In some embodiments, the bezel **170** is disposed under the opening portion **123** and includes a flexible material. For example, the bezel **170** may include polymer or any other suitable organic material. When the cover plate **110** is in the closed state relative to the base **120** (for example, there is a 0° angle between the cover plate **110** and the base **120**), since the cover plate **110** substantially covers the top surface **122** of the base **120**, the portion of the bezel **170** (for example, the portion connected to the cover plate **100**) may extend above the top surface **122** of the base **120**.

(27) FIG. 9 illustrates a cross-sectional view of the electronic device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 9, when the cover plate **110** is in an open state relative to the base **120** (for example, there is a 90° angle between the cover plate **110** and the base **120**), the bezel **170** extends from the bottom of the opening portion **123** to the cover plate **110**. The bezel **170** may extend to completely cover the gap between the opening portion **123** and the cover plate **110**, so that the hot air (for example, the airflow F) discharged from the opening portion **123** after the heat exchange would not flow back into the base **120**, enhancing the heat dissipation efficiency of the electronic device **100**.

(28) In some embodiments, the bezel **170** (for example, the first end **170A**) may be fixed to the cover plate **110** and is movable relative to the base **120** (for example, the second end **170B**), but the present disclosure is not limited thereto. In other embodiments, the first end **170A** of the bezel **170** may be fixed to the cover plate **110**, and the second end **170B** may be fixed to the base **120**. Since the bezel **170** is made of a flexible material, it can expand or contract along with the rotation of the cover plate **110** and will not be damaged due to the change of the position of the cover plate **110**.

(29) As set forth above, the present disclosure provides an electronic device including a bezel. The arrangement of the bezel may reduce the distance (possibly down to 0) between the bezel and the down edge of the opening portion of the base, further reduce the airflow after the heat exchange flowing downwards and back to the heat dissipation mechanism in the base. The heat dissipation efficiency of the electronic device may be improved. In addition, the bezel may also be combined with a sensor structure that is commonly used in electronic devices to improve the accuracy of the operation of the bezel.

(30) While the embodiments and the advantages of the present disclosure have been described above, it should be understood that those skilled in the art may make various changes, substitutions, and alterations to the present disclosure without departing from the spirit and scope of the present disclosure. In addition, the scope of the present disclosure is not limited to the processes, machines, manufacture, composition, devices, methods and steps in the specific embodiments described in the specification. Those skilled in the art may understand existing or developing processes, machines, manufacture, compositions, devices, methods and steps from some embodiments of the present disclosure. As long as those may perform substantially the same function in the aforementioned embodiments and obtain substantially the same result, they may be used in accordance with some embodiments of the present disclosure. Therefore, the scope of the present disclosure includes the aforementioned processes, machines, manufacture, composition, devices, methods, and steps. Furthermore, each of the appended claims constructs an individual embodiment, and the scope of the present disclosure also includes every combination of the appended claims and embodiments.

Claims

1. An electronic device, comprising: a base having a sidewall, wherein an opening portion is formed on the sidewall, and an airflow selectively passes through the opening portion; a cover plate

pivotally connected to the base; a bezel having a longitudinal top edge and a longitudinal bottom edge, the longitudinal top edge pivotally connected to the cover plate; and a driving assembly, a part of the driving assembly located in the base and another part of the driving assembly disposed on the cover plate and abutting with the bezel; wherein when cover plate is rotated from a closed position to an open position, the driving assembly drives the bezel to rotate relative to the cover plate such that the longitudinal bottom edge is closer to the sidewall than the longitudinal top edge, and the bezel deflects the airflow passing out of the opening portion.

2. The electronic device as claimed in claim 1, wherein the bezel is pushed to rotate via the driving assembly selectively.

3. The electronic device as claimed in claim 2, wherein the driving assembly comprises: a cam selectively rotating with the rotation of the cover plate relative to the base; and a push rod connected to the cam, and one end of the push rod is abutted against the bezel.

4. The electronic device as claimed in claim 3, wherein the push rod comprises an engaging portion, the engaging portion is provided with a plurality of teeth, and the teeth of the engaging portion are engaged with a plurality of teeth of the cam.

5. The electronic device as claimed in claim 4, wherein the cam and the engaging portion have different numbers of teeth.

6. The electronic device as claimed in claim 4, wherein the engaging portion has a fan-shaped gear structure.

7. The electronic device as claimed in claim 1, wherein when the cover plate is opened relatively to the base, a first gap is formed between the bezel and an up edge of the opening portion, a second gap is formed between the bezel and the down edge of the opening portion, and the first gap is greater than the second gap.

8. The electronic device as claimed in claim 7, wherein the first gap is greater than five times of the second gap.

9. The electronic device as claimed in claim 7, wherein when the cover plate is opened relatively to the base, a distance from an up edge of the bezel to a horizontal plane is greater than a distance from the up edge of the opening portion to the horizontal plane.

10. The electronic device as claimed in claim 7, wherein when the cover plate is opened relatively to the base, a distance from a down edge of the bezel to a horizontal plane is less than a distance from the down edge of the opening portion to the horizontal plane.

11. An electronic device, comprising: a base having a sidewall, wherein an opening portion is formed on the sidewall, and an airflow selectively passes through the opening portion; a cover plate pivotally connected to the base; a bezel having a longitudinal top edge and a longitudinal bottom edge, the longitudinal top edge pivotally connected to the cover plate; and a driving assembly connected to the cover plate, wherein the bezel is pushed to rotate via the driving assembly selectively, wherein the driving assembly comprises: a motor disposed on a surface of the cover plate facing the base and selectively driving the bezel to rotate relative to the cover plate such that the longitudinal bottom edge is closer to the sidewall than the longitudinal top edge, and the bezel deflects the airflow passing out of the opening portion.

12. The electronic device as claimed in claim 11, wherein the driving assembly further comprises a sensor, a relative position of the cover plate and the base is detected by the sensor, and the bezel is selectively driven by the motor to rotate via detecting the relative position for controlling the motor.

13. The electronic device as claimed in claim 11, wherein the bezel is pivotally connected to a pivot rod of the cover plate via a linkage portion, and the motor is disposed adjacent to the linkage portion.

14. The electronic device as claimed in claim 13, wherein the motor is connected to the pivot rod, the linkage portion is selectively driven by the motor via the pivot rod, the bezel is driven to rotate toward the opening portion via the linkage portion.

15. The electronic device as claimed in claim 13, when viewed in a direction parallel to a rotation

axis of the bezel, the motor overlaps with the pivot rod.

16. An electronic device, comprising: a base having a sidewall, wherein an opening portion is formed on the sidewall, and an airflow selectively passes through the opening portion; a cover plate pivotally connected to the base; a bezel having a longitudinal top edge and a longitudinal bottom edge, the longitudinal top edge pivotally connected to the base and the cover plate, wherein the bezel is disposed below the opening portion; and a driving assembly, a part of the driving assembly located in the base and another part of the driving assembly disposed on the cover plate and abut with the bezel; wherein when cover plate is rotated from a closed position to an open position, the driving assembly drives the bezel to rotate relative to the cover plate such that the longitudinal bottom edge is closer to the sidewall than the longitudinal top edge, and the bezel deflects the airflow passing out of the opening portion.

17. The electronic device as claimed in claim 16, wherein the bezel comprises a flexible material.

18. The electronic device as claimed in claim 16, wherein when the cover plate is closed relatively to the base, a portion of the bezel extends over a top surface of the base.

19. The electronic device as claimed in claim 16, wherein when the cover plate is opened relatively to the base, the bezel extends from a bottom of the opening portion toward the cover plate.

20. The electronic device as claimed in claim 16, wherein the bezel is fixed to the cover plate and is movable relatively to the base.
